Laplace Transformation

Assignment 5 — Six-Page Study Guide

Basic ideas, formulas, the six original worked questions, and practice with solutions

1. What is a Laplace transform?

A Laplace transform changes a function of *time*, $f(t)$, into a function of a new variable s, written $F(s)$. It is useful because operations such as differentiation and integration in time can become algebraic operations in s. In this assignment, we mainly use standard formulas and three properties instead of evaluating every integral from scratch.

$$\boxed{F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt}$$

Symbols: t is time; s is a complex variable; e is the exponential base; $f(t)$ is the input; $F(s)$ is its transform. The formula applies where the integral converges. For ordinary exam examples, the relevant range of s is assumed.

Convention. The handwritten first page displays bounds $-\infty$ to ∞ (the *two-sided* transform), but its calculations use 0 to ∞. This guide follows those calculations: the *one-sided* Laplace transform.

2. Essential direct-transform formulas

Input $f(t)$	Transform $F(s)$
Constant k	$\dfrac{k}{s}$
Ramp t	$\dfrac{1}{s^2}$
Exponential e^{at}	$\dfrac{1}{s-a}$
Sine $\sin(at)$	$\dfrac{a}{s^2+a^2}$
Cosine $\cos(at)$	$\dfrac{s}{s^2+a^2}$

Here k and a are constants. For example, $\mathcal{L}\{e^{3t}\} = 1/(s-3)$, while $\mathcal{L}\{e^{-7t}\} = 1/(s+7)$. The minus sign in $s-a$ matters.

Question 1 (Assignment p. 1)

Original question: "Laplace Transform of ramp func, $x(t) = t$." Find $\mathcal{L}\{t\}$.

Step 1. Substitute $f(t) = t$ into the definition:

$$\mathcal{L}\{t\} = \int_0^\infty te^{-st}\,dt.$$

Step 2. Integrate by parts ($\int u\,dv = uv - \int v\,du$). Take $u = t$ and $dv = e^{-st}dt$; then $du = dt$ and $v = -e^{-st}/s$:

$$\int te^{-st}\,dt = -\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}.$$

Step 3. Evaluate at ∞ and 0. For $s > 0$ the terms vanish at infinity; at 0 the value is $-1/s^2$:

$$\mathcal{L}\{t\} = \left[-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right]_0^\infty = 0 - \left(-\frac{1}{s^2}\right).$$

$$\boxed{\mathcal{L}\{t\} = \frac{1}{s^2}}$$

Remember: Once this standard result is known, use it directly in later questions.

3. First shifting property

When to use it: an exponential e^{at} multiplies another function $f(t)$.

$$\boxed{\mathcal{L}\{e^{at} f(t)\} = F(s-a)}, \qquad F(s) = \mathcal{L}\{f(t)\}.$$

Meaning: First transform $f(t)$ *without* the exponential. Then replace *every* occurrence of s with $s - a$.

Why it works: In the defining integral, $e^{at}e^{-st} = e^{-(s-a)t}$. This is exactly the original transform with s replaced by $s - a$.

Question 2 (Assignment p. 2)

Original question: $\mathcal{L}[e^{5t} \cdot t]$.

Step 1. Identify the ordinary function: $f(t) = t$. Its direct transform is

$$F(s) = \mathcal{L}\{t\} = \frac{1}{s^2}.$$

Step 2. Identify the exponential exponent: e^{5t} gives $a = 5$.

Step 3. Make the replacement $s \mapsto s - 5$:

$$F(s-5) = \frac{1}{(s-5)^2}.$$

Step 4. If an expanded denominator is requested, use $(s-5)^2 = s^2 - 10s + 25$.

$$\boxed{\mathcal{L}[e^{5t} \cdot t] = \frac{1}{(s-5)^2} = \frac{1}{s^2 - 10s + 25}}$$

Question 3 (Assignment p. 3)

Original question: $\mathcal{L}[e^{-2t}t]$.

Step 1. Ignore the exponential first: $\mathcal{L}\{t\} = 1/s^2$.

Step 2. Here $a = -2$, so the shift is

$$s - a = s - (-2) = s + 2.$$

Step 3. Replace s everywhere by $s + 2$:

$$F(s+2) = \frac{1}{(s+2)^2}.$$

Step 4. Expand only if needed: $(s+2)^2 = s^2 + 4s + 4$.

$$\boxed{\mathcal{L}[e^{-2t}t] = \frac{1}{(s+2)^2} = \frac{1}{s^2 + 4s + 4}}$$

Quick sign test

Factor	a	Replace s with
e^{5t}	$+5$	$s - 5$
e^{-2t}	-2	$s + 2$
e^{-7t}	-7	$s + 7$

Common error: Replacing s by $s - 2$ for e^{-2t}. The two minus signs make a plus: $s - (-2) = s + 2$.

4. Multiplication by t property

When to use it: the original function contains $t\,f(t)$, where the transform of $f(t)$ is known.

$$\boxed{\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}}, \qquad F(s) = \mathcal{L}\{f(t)\}\,.$$

This means: (1) ignore the multiplying t temporarily; (2) find $F(s)$; (3) differentiate it with respect to s; (4) put a minus sign in front.

Why it works: Differentiating the exponential in the integral gives $\frac{d}{ds}(e^{-st}) = -te^{-st}$. Hence $F'(s) = -\mathcal{L}\{tf(t)\}$ when differentiation under the integral is valid.

Question 4 (Assignment p. 3)

Original question: $\mathcal{L}[t \cdot \sin 4t]$.

Step 1. Ignore t. Use the sine transform with $a = 4$:

$$F(s) = \mathcal{L}\{\sin 4t\} = \frac{4}{s^2 + 4^2} = \frac{4}{s^2 + 16}.$$

Step 2. Apply the multiplication property:

$$\mathcal{L}\{t \sin 4t\} = -\frac{d}{ds}\left(\frac{4}{s^2 + 16}\right).$$

Step 3. Differentiate. For a fraction, use the quotient rule:

$$\frac{d}{ds}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}.$$

Take $u = 4$, $u' = 0$; $v = s^2 + 16$, $v' = 2s$. Therefore,

$$\frac{d}{ds}\left(\frac{4}{s^2 + 16}\right) = \frac{(s^2 + 16)(0) - 4(2s)}{(s^2 + 16)^2} = -\frac{8s}{(s^2 + 16)^2}.$$

Step 4. Apply the *extra minus sign* from the property:

$$-\left(-\frac{8s}{(s^2 + 16)^2}\right) = \frac{8s}{(s^2 + 16)^2}.$$

$$\boxed{\mathcal{L}[t \sin 4t] = \frac{8s}{(s^2 + 16)^2}}$$

Equivalent expanded answer: $\dfrac{8s}{s^4 + 32s^2 + 256}$. Leave the denominator factored unless expansion is required.

Another form to recognize

For $t^n f(t)$, the property becomes

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}}, \qquad n = 1, 2, 3, \ldots$$

For $t^2 f(t)$, differentiate *twice*. Because $(-1)^2 = +1$, there is no final minus sign. This is the same rule, not a separate topic.

Check before moving on: If you start with $\mathcal{L}\{\sin(at)\} = a/(s^2 + a^2)$, this method gives $\mathcal{L}\{t \sin(at)\} = 2as/(s^2 + a^2)^2$. Setting $a = 4$ produces $8s/(s^2 + 16)^2$.

5. Combining properties: the main worked problem

Most exam questions combine two or more properties. **Use this order for the next question:** direct transform → multiplication by t → first shifting.

Question 5 (Assignment p. 4–5)

Original question:

$$\boxed{\mathcal{L}[\cos 2t \cdot t \cdot e^{-5t}]}.$$

The three pieces tell you which operations to use:

Part of the function	Operation
$\cos 2t$	Direct cosine transform
t	Differentiate the transform and negate
e^{-5t}	Replace s with $s + 5$

Step 1: Transform $\cos 2t$.. The cosine formula gives

$$F(s) = \mathcal{L}\{\cos 2t\} = \frac{s}{s^2 + 4}.$$

Step 2: Account for the factor t.. Apply the multiplication property:

$$\mathcal{L}\{t \cos 2t\} = -\frac{d}{ds}\left(\frac{s}{s^2 + 4}\right).$$

Using the quotient rule with $u = s$, $u' = 1$, $v = s^2 + 4$, $v' = 2s$:

$$\begin{aligned} -\frac{d}{ds}\left(\frac{s}{s^2 + 4}\right) &= -\frac{(s^2 + 4)(1) - s(2s)}{(s^2 + 4)^2} \\ &= -\frac{s^2 + 4 - 2s^2}{(s^2 + 4)^2} = \frac{s^2 - 4}{(s^2 + 4)^2}. \end{aligned}$$

Step 3: Account for e^{-5t}.. The shift has $a = -5$, so

$$s \mapsto s - (-5) = s + 5.$$

Apply the substitution to *both* numerator and denominator:

$$\mathcal{L}[te^{-5t} \cos 2t] = \frac{(s + 5)^2 - 4}{\left((s + 5)^2 + 4\right)^2}.$$

Step 4: Expand only inside the brackets..

$$(s + 5)^2 - 4 = s^2 + 10s + 21, \qquad (s + 5)^2 + 4 = s^2 + 10s + 29.$$

$$\boxed{\mathcal{L}[\cos 2t \cdot t \cdot e^{-5t}] = \frac{s^2 + 10s + 21}{(s^2 + 10s + 29)^2}}$$

Important: The entire denominator is squared. Do *not* multiply the separate transforms of t, e^{-5t}, and $\cos 2t$ together; in general, $\mathcal{L}\{fg\} \neq \mathcal{L}\{f\}\ \mathcal{L}\{g\}$.

6. Integration property: the final assignment question

When to use it: the question asks for the Laplace transform of an integral *from* 0 *to* t.

$$\boxed{\mathcal{L}\left\{\int_0^t f(u)\,du\right\} = \frac{F(s)}{s}}, \qquad F(s) = \mathcal{L}\{f(t)\}\,.$$

The u is a *dummy variable* inside the integral; t is its upper limit. Using two letters prevents confusion.

Why this works: Write $g(t) = \int_0^t f(u)\,du$. Then $g'(t) = f(t)$ and $g(0) = 0$. The derivative rule $\mathcal{L}\{g'(t)\} = sG(s) - g(0)$ gives $F(s) = sG(s)$, so $G(s) = F(s)/s$.

Question 6 (Assignment p. 5)

Original handwritten question (notation preserved):

$$\mathcal{L}\left[\int_0^t \cos 2t \cdot t \cdot e^{-5t}\,dt\right].$$

The intended expression, written unambiguously with u as the integration variable, is

$$\boxed{\mathcal{L}\left\{\int_0^t ue^{-5u}\cos(2u)\,du\right\}}.$$

Step 1. Identify the integrand: $f(u) = ue^{-5u}\cos(2u)$. Its transform has already been worked out in Question 5:

$$F(s) = \frac{s^2 + 10s + 21}{(s^2 + 10s + 29)^2}.$$

Step 2. Apply the integration property. *Divide the entire transform by s:*

$$\mathcal{L}\left\{\int_0^t ue^{-5u}\cos(2u)\,du\right\} = \frac{1}{s}\left(\frac{s^2 + 10s + 21}{(s^2 + 10s + 29)^2}\right).$$

Step 3. Combine the denominators. The numerator remains unchanged.

$$\boxed{\mathcal{L}\left\{\int_0^t ue^{-5u}\cos(2u)\,du\right\} = \frac{s^2 + 10s + 21}{s(s^2 + 10s + 29)^2}}$$

One idea ties the last two questions together

Input	Transform
$te^{-5t}\cos 2t$	$\dfrac{s^2 + 10s + 21}{(s^2 + 10s + 29)^2}$
$\displaystyle\int_0^t ue^{-5u}\cos 2u\,du$	$\dfrac{s^2 + 10s + 21}{s(s^2 + 10s + 29)^2}$

Difference: The integral in the second row adds precisely one factor s to the denominator. There is no need to evaluate the time-domain integral.

7. One-page revision and self-test

Which operation do you use?

Pattern in question	Action in the s-domain
k, t, e^{at}, $\sin(at)$, $\cos(at)$	Use the standard formula.
$e^{at}f(t)$	Shift: replace every s by $s-a$.
$tf(t)$	Differentiate $F(s)$ and put a minus sign.
$t^n f(t)$	Differentiate n times and multiply by $(-1)^n$.
$\int_0^t f(u)\,du$	Divide $F(s)$ by s.

Standard transforms — memorise these five

$$\boxed{\begin{aligned}&\mathcal{L}\{k\}=\frac{k}{s}, \qquad \mathcal{L}\{t\}=\frac{1}{s^2}, \qquad \mathcal{L}\{e^{at}\}=\frac{1}{s-a},\\ &\mathcal{L}\{\sin(at)\}=\frac{a}{s^2+a^2}, \qquad \mathcal{L}\{\cos(at)\}=\frac{s}{s^2+a^2}.\end{aligned}}$$

Linearity (useful for sums): $\mathcal{L}\{Af(t)+Bg(t)\}=AF(s)+BG(s)$ for constants A, B.

Practice A — direct transform

Question: Find $\mathcal{L}\{4t+3+\sin 2t\}$.

Solution: Transform each term separately:

$$\mathcal{L}\{4t+3+\sin 2t\}=4\mathcal{L}\{t\}+3\mathcal{L}\{1\}+\mathcal{L}\{\sin 2t\}=\boxed{\frac{4}{s^2}+\frac{3}{s}+\frac{2}{s^2+4}}.$$

Practice B — two properties together

Question: Find $\mathcal{L}\{te^{-3t}\sin 2t\}$.

Solution: Direct: $\mathcal{L}\{\sin 2t\}=2/(s^2+4)$. Multiply by t: $-\frac{d}{ds}[2/(s^2+4)]=4s/(s^2+4)^2$. Shift for e^{-3t}: $s\mapsto s+3$.

$$\boxed{\mathcal{L}\{te^{-3t}\sin 2t\}=\frac{4(s+3)}{((s+3)^2+4)^2}}.$$

Practice C — integration

Question: Find $\mathcal{L}\left\{\int_0^t \cos(2u)\,du\right\}$.

Solution: First $\mathcal{L}\{\cos 2t\}=s/(s^2+4)$. Divide by s:

$$\boxed{\mathcal{L}\left\{\int_0^t \cos(2u)\,du\right\}=\frac{1}{s}\frac{s}{s^2+4}=\frac{1}{s^2+4}}.$$

Final exam checklist

1. Underline the *base* function, the factor t, the factor e^{at}, and any integral.
2. Transform the base function; apply multiplication by t, then shifting, then integration as needed.
3. Check the sign of a, replace *all* occurrences of s, and retain full denominator brackets.
4. Leave the final answer factored unless expansion is specifically requested.

Source: IDSA Assignment 5, five handwritten pages. Questions 1–6 reproduce its worked problem statements (with a clearly marked notation clarification in Question 6). The practice problems on this page are additional exercises.